

System audit, 9 October 2026

What the system proves, what is wrong in it, what was corrected during the audit, and the order in which the rest should be done

9 October 2026

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Scope is the MAICC-IFRS9 system at commit f5b08fc on branch eir_revenue_recognition, audited against specification v4 of 7 October 2026, IFRS 9, the RBM Credit Risk Management for DFIs Directive 2018, and ordinary standards of security, data integrity and operability. Confidential to MAIIC and Dupleix.

1. Verdict

The system does what the After Build Report of 8 October says it does in the sense that every step named there runs from one command on a wiped database and produces the figures quoted. The audit went behind those figures, and the honest summary is this: **the data foundation is sound and proven to the cent; the impairment engine is not yet on an IFRS 9 basis; and the legacy modules the platform was forked from expose the whole loan book and its governed inputs without a login.** None of the three can be left as it is before MAIIC relies on a number from the system or puts it on a network.

In order of consequence:

- 1 **The ECL is a 12-month ECL for every stage.** Stage 2 loans (8.35bn of exposure, 36 percent of the book at August 2026) are provided at the 12-month PD, not the lifetime PD; the lifetime PD the engine computes is written to a column nothing reads. The allowance is also undiscounted in the bootstrap. The 8.94bn quoted in the After Build Report understates a lifetime, discounted ECL by an amount that cannot be stated without running the correct engine, but a Stage 2 loan with 36 months to run and a 10 percent annual PD is provided at 10 percent where IFRS 9 requires 27.
- 2 **Stage 3 interest is recognised on the gross amount,** not the net carrying amount, because the revenue engine reads an allowance column (expected_loss_provision) that the governed build never writes. The "net basis" setting is in force and the code path exists; it has no allowance to net. The test that covers it seeds the column by hand.
- 3 **Write-offs (type 300) and the Nascomex equity entries (type 401) are treated as cash received** by the loan-book build and the EIR roll-forward, against the specification's own list at section 3.3 (22.3m and 535.7m in the committed pack).
- 4 **180 web routes carry no authentication,** including the loan-book import, the collateral register, the transition-matrix and LGD writes into the loan book, the ECL recalculation and a one-click regression approval. One of them interpolates request input into a raw SQL statement. All of these are modules of the product the platform was forked from, not work of this engagement, but they are on the server MAIIC will run.
- 5 **The RBM classification bands built on 8 October were wrong for medium- and long-term facilities** (they put a 31-to-90-day facility in special mention at 5 percent where the Gazette says standard at 0, and used 360/720 where the Gazette says 365/746). This was Dupleix's error, found by this audit and corrected the same day at commit f5b08fc; the test now asserts every boundary in the Gazette.

The rest of this report gives the evidence, the full list of findings by severity, what was corrected during the audit, and the order in which the remainder should be done.

2. How the audit was done

Everything below rests on things that were run or read, not on the After Build Report's own claims.

Proof	Result
Clean install: <code>eir:bootstrap --fresh --force-wipe --with-client-inputs --build --run-engines --verify</code> on a wiped throwaway database	Completed in 183 seconds; 9 of 9 golden checks pass (ledger interest 5,293,988,207.06; method B equals the stored report on 2,361 of 2,361 account-months; the December 2025 GL ties incl. the two accepted FInES exceptions; the 4215/4216 year-end legs).
Every IFRS 9 report rendered for August 2026 (<code>ifrs9:smoke-reports</code> , now all 30 methods)	All OK after the audit's own fixes (two reports had thrown a division error on a sector with no exposure; see section 6).
Full test suite	See section 7 for the counts; the suite cannot complete in one process under XAMPP's 300-second execution limit, which is itself a finding.
Dependency audit (<code>composer audit, npm audit --omit=dev</code>)	Composer: two medium advisories (symfony/routing CVE-2026-48784 and CVE-2026-45065), one low (symfony/polyfill-intl-idn), two abandoned packages (beyondcode/laravel-websockets, box/spout). npm: 33 advisories (1 critical, 19 high) led by TinyMCE XSS in the help editor and a yaml stack overflow.
Route inventory (<code>route:list</code> with controller middleware merged) and a read of every <code>\$this->middleware</code> in the controllers	180 of 724 web routes have no auth middleware.
Four independent code reviews, each read-only with file-and-line evidence: specification conformance (sections 3 to 16, every decision, the 48-setting catalogue), financial-engine correctness, security and access control, data integrity and operability	The reviews' findings were re-read against the code before inclusion; the Critical items in section 4 were each re-verified by hand.
The Gazette itself (<code>docs/regulatory</code> , pages 683 and 685)	Read in full for the classification bands and rates.
The bootstrapped database queried directly for coverage	135 loans in the August 2026 book; 144 contracts in the EIR master; 31 schedules generated and 12 EIRs locked (section 5).

3. What the system proves (and should keep)

- **The ledger is the record and the build reproduces it.** Method B derives every account-month's carrying amount from the landed ledger and equals E-Banker's own stored report to the cent on 2,361 of 2,361 account-months; the loan GLs tie to the December 2025 trial balance to the cent, with the two FInES exceptions carried as accepted exceptions rather than hidden.
- **Landing, gates and versioning.** Packs land with hashes, counts, dates, key uniqueness and query-id checks; rows are append-only and versioned with their pack; a failed gate quarantines the load.
- **Maker-checker is enforced server-side** on every path built in this engagement (loan-book build, governance settings, scenario sets, regression fits, compliance sign-off); the bootstrap's own approvals carry a label that is visibly not a person's.
- **The EIR solver** is sound: Newton with bisection fallback, bounded, residual re-checked; locked EIRs cannot be recomputed; reopening archives and voids downstream rows.
- **Staging** reads days past due from the oldest overdue instalment under the governed basis, applies the directive's Stage 3 thresholds by tenor class (91 days short-term, 181 otherwise), allocates receipts to instalments in due-date order for the four-instalment trigger, and writes the post-qualitative stage the ECL and FLI read.
- **Units are consistent where it matters most:** the facility utilisation rate is stored as a fraction and read as one by every EAD reader; PD is stored in percent in the matrix and divided into a fraction by both PD writers; Stage 3 PD is forced to 1 in every engine.

- **The forward-looking chain** fails closed (observations, sign, R-squared, p-value), declines with reasons, and the post-FLI PD is bounded and weighted by validated weights.
- **The suite shell and theme**: every one of 120 captured screens renders in dark mode (measured, not assumed) and the light theme is unchanged.
- **Governance of the feed token**: the API route compares the bearer token in constant time and refuses an empty configured token.

4. Findings

Severity: **Critical** = a wrong number in a financial statement or a regulatory return, or an unauthenticated path to financial data; **High** = wrong under a realistic condition, or a control that can be bypassed; **Medium** = a fragile or undocumented assumption, or a gap against the specification that does not yet move a number; **Low** = cosmetic or latent. Each finding names the file so an engineer can go straight to it.

4.1 Critical

#	Finding	Where	Effect and example	Fix
C1	Stage 2 allowance is a 12-month ECL; <code>lifetime_pd</code> is written and never read; the bootstrap books an undiscounted ECL	<code>ExpectedCreditLossController.php:375-383</code> ($ecl = pd \times lgd \times EAD$ with the 12-month PD for every stage); <code>RecalculateEcl.php:24</code> defaults <code>--discounting=undiscounted</code> and <code>Bootstrap.php:315</code> accepts the default; <code>PdEngineService.php:71</code> writes <code>lifetime_pd</code>	Stage 2, 12m PD 10%, 36 months remaining, LGD 0.5, EAD 1,000,000: booked 50,000; $lifetime\ 1 - 0.9^3 = 27.1\%$ gives 135,500 before discounting. Every report, the FS note and the RBM return read <code>ec1_value</code>	Stage 2 uses <code>lifetime_pd</code> ; the bootstrap runs the time-phased engine (<code>--discounting=discounted</code>) once C2 and C3 are fixed; the FS note then carries lifetime, discounted figures
C2	<code>remaining_tenor</code> is months from the governed build and years from the legacy importer; readers disagree	Writes months: <code>LoanBookBuildService.php:549</code> . Writes years: <code>ProcessLoanImportJob.php</code> , <code>LoanBooksImport.php</code> . Reads months: <code>PdEngineService.php:70</code> . Reads years: <code>TransitionMatrixController.php:523, 546</code> ($lifetime\ PD = 1 - (1 - pd)^{tenor}$), <code>TimePhasedEclService.php:169</code> ($\times 12$), <code>EclDiscountingService.php:81</code>	Under build B a 24-month Stage 2 loan: the Monthly Probability screen's lifetime PD is $1 - 0.9^{24} = 92\%$ against the engine's 19%; the discounting service discounts over 24 "years"	One unit (months), stated on the column, and every reader converted; a test per reader
C3	Stage 3 ECL in the time-phased engine is discounted from the wrong date and the recovery plan is netted before discounting	<code>TimePhasedEclService.php:69, 105, 114-115</code>	EAD 100,000, LGD 0.6, 24-month schedule, EIR 10%: 49,587 instead of 60,000; the covering test asserts the understated range	Loss at the reporting date, or EAD less the present value of each recovery from its own date; the LGD fed in is the undiscounted cohort LGD (<code>LgdEngineService.php:70</code>) and should be discounted
C4	RBM classification bands wrong for medium- and long-term facilities (built 8 October)	<code>RbmReturnService.php:36</code> , <code>Ifrs9ReportsController.php</code> (RBM constant and <code>rbmClassCase</code>), <code>tests/Feature/Rbm/RbmReturnServiceTest.php</code>	A 36-month facility 60 days overdue on 10m: minimum provision 500,000 instead of 0; 361-365 days long-term Doubtful (50%) not Substandard (20%); 721-746 Loss not Doubtful	Corrected at f5b08fc to the Gazette's section 10 (short-term 30/90/180/365; medium and long 90/180/365/746); the test asserts every boundary on both terms; workbook 4 F3 records both corrections
C5	The revenue roll-forward deducts scheduled instalments as cash after the borrower's last receipt	<code>EirRevenueService.php:171-178</code> : "covered" is the contract's own first and last transaction; after the last receipt every month falls back to the schedule	Loan 1,000 at 1% a month, 100 a month schedule, last receipt March: by December the engine's gross is about 100 while the ledger shows about 1,100. This is the Stage 3 population	The window is the feed's coverage (<code>LandingZoneReader::lastLedgerDate()</code>), not the contract's; a month inside coverage with no receipt is zero cash

#	Finding	Where	Effect and example	Fix
C6	Stage 3 "net carrying amount" interest never takes effect	EirRevenueService.php:70 reads loan_books.expected_loss_provision; the governed build's column list (LoanBookBuildService.php:45-53) does not write it; only the legacy LoanBookController.php:331 and test data do. The bootstrap also runs revenue before the ECL (Bootstrap.php:226-258 then :315)	Gross 1,000, allowance 600, 1% a month: recognised 10, should be 4, on every Stage 3 loan (8.26bn of exposure)	Read ecl_value (the allowance the engine writes); run the ECL for the prior period before the revenue step, or carry the opening allowance on the roll-forward row
C7	180 web routes have no authentication	routes/web.php modules: loss-given-default (22 routes), transition-matrix (19), loan_application (18), collateral (13), credit-loss-data (11), transition-matrix-cummulative (10), lgd-calculations (9), regression (9), file (9), expected-credit-loss (5), groups, scenarios, scenario-profiles, macro-forecast-weighted, macro-statistics (legacy), transition-profile, api/*	POST /transition-matrix/{id}/update-loan-book rewrites the PDs that feed the ECL; POST /expected-credit-loss/calculation recomputes the allowance; GET /api/collateral-registers returns the whole collateral register as JSON with CORS open to any origin; PATCH /regression/{model}/approve approves a model with no user and no log. None needs a cookie	Wrap every legacy module in auth plus a permission; delete the dead routes/api.php routes (two name controller methods that do not exist); this is a day's work and the first thing to do before the server is reachable from the MAIIC network
C8	Unauthenticated SQL interpolation	TransitionProfileOptionController.php:66-96 (POST /transition-profile/categories/reorder): categories.*.id validated with exists: then interpolated into DB::statement("... WHEN {\$id} THEN ..."). MySQL's numeric coercion lets "5 OR 1=1" pass an exists check against id 5	Arbitrary SQL in the CASE expression from an unauthenticated request	Bind the ids; cast to int; authenticate the route (C7)
C9	The bootstrap approves a regression fit under its own label and computes the ECL on it	Bootstrap.php:298-309 proposes, approves and applies the best fit; spec 6.10.1 step 6 says the overlay is zero and the post-FI PD marked "no adjustment: bootstrap" until two people approve a fit; the impairment workbook's own finding F2 states the spec's rule	The 8.94bn ECL (and the 10.24bn weighted sensitivity) in the After Build Report rests on a fit no MAIIC person approved; FliRouteService.php:52 lets any non-null label bypass the different-person check	The bootstrap applies the route with the overlay at zero and marks the PD; fit approval is a screen action by two people; the label bypass is limited to the bootstrap's constant

4.2 High

#	Finding	Where	Effect	Fix
H1	Write-offs (300) and the Nascomex equity entries (401) are classed as cash receipts	LoanBookBuildService.php:42 (TYPE_RECEIPT includes 401 and 300); ContractInputsBuildService.php:176-180	22.3m of write-offs and 535.7m of equity entries in the committed pack flow into repayments and the EIR roll-forward's cash_received. Spec 3.3 lists 300 as write-off and 400/401 as equity. The method B tie to the stored report still passes because the stored report makes the same mistake; the tie proves reproduction, not correctness	A write-off reduces gross carrying through the allowance, not cash; 401 is excluded from loan cash; re-prove the ties with the exceptions stated
H2	Tranche facilities: later drawdowns never enter the EIR cash-flow vector	EirContractInputService.php:81-82 uses drawn_amount at origination and the version-1 schedule; DisbursementService::laterDrawdownFlows is unused; EirReadinessService.php:30,44-49	Approved 100m, 60m drawn at origination and 40m six months later: the solver sees 60m out and 100m plus interest in; the EIR solves far above the contractual rate. Spec v4 counts 58 tranching facilities	Negative (disbursement) flows at their dates from P3_12; a test on a two-tranche facility
H3	Scenario weighting is applied twice over two different sets	FlirouteService.php:127-136 writes the probability-weighted PD across the governed set; TimePhasedEclService.php:82-85 then weights again by ecl_scenario_assumptions (a seeded "controlled synthetic" set, BASE 60 / UPSIDE 20 / DOWNSIDE 20), which Bootstrap.php:105 seeds as approved	Double-weighting whenever the time-phased engine runs; and two different per-scenario answers for the same period (the sensitivity's typed multipliers against the route's regression adjustments: 10.24bn against 8.94bn in the After Build Report)	One governed set; the ECL per scenario, weighted once, and the sensitivity reads the same per-scenario ECLs (spec 15.5)
H4	The reporting-period lock is honoured only by the loan-book build	LoanBookBuildService.php:616 is the sole reader of reporting_period_locks; EirRevenueService::run (recalculate: true) supersedes rows in a locked period; ECL, staging, PD, LGD and FLI write freely	A locked month can be restated by any engine; EirAsAtService.php:66-69 claims the opposite	One lock check in a shared helper, called by every writer; a test per writer
H5	Reports and the FS note split by the pre-qualitative stage; the ECL was measured on the post-qualitative stage	Ifrs9ReportsController.php:137,161,268,301,333,367,390,419,628; ExpectedCreditLossController.php:420; StressTestService.php:25-26,44	A loan moved to Stage 3 by the four-installment trigger (post-only) appears in "Stage 1" with PD 1.0 in "Closing ECL by Stage" and the impairment charge by stage	Every reader splits by ifrs9 stage_post_qualitative
H6	The FS note's "gross carrying amount" is the EAD	Ifrs9ReportsController.php:628-633 uses carrying + commitments × utilisation as gross carrying; NPL ratios at :573 and :1240 are on EAD	Carrying 60m, commitments 40m, utilisation 0.6: "gross carrying" 84m in the note	Gross carrying = carrying amount; EAD is its own column

#	Finding	Where	Effect	Fix
H7	LGD cohort workout counts a written-off loan as fully recovered and double-counts a cured loan that also paid down; recoveries undiscounted; no collateral	LgdEngineService.php:38,49-56,70	Cohort of two 500s, one written off (absent from the end book): recovery 50 percent, LGD 0.5; the true recovery on the absent loan is 0	Absent-from-book is a write-off unless the ledger shows settlement; cure and recovery are exclusive; discount at the EIR; collateral from the register
H8	12-month PD from a short window is not annualised	PdEngineService.php:36-38,65-67	With books from October 2025 run for January 2026, the "12-month PD" is a three-month default rate	Annualise $1-(1-p)^{(12/\text{months})}$ and record the window
H9	The admin role bypasses maker-checker on governance approvals and EIR locks	EirGovernanceController.php:67-70 (adminOverride() = has role admin), used at ten call sites; EirCalculationService.php:108-111	An administrator can propose and approve a governed setting alone, and lock an EIR they calculated	The override is a governed setting with its own approval, or removed; schedule approval (ScheduleWorkflowService.php:275) should also require a different person from the editor
H10	Governance defaults are effective from 1 January 2025; 63 of 144 contracts originated before 2024	GovernanceSettingsSeeder.php:27 (EFFECTIVE_FROM); GovernanceService::inForce() resolves at the contract's origination date	16 contracts on the clean install cannot generate a schedule because "no approved value is in force for day_count" at origination; the other 47 pre-2024 contracts fail earlier for other reasons and would hit this next	Seeded defaults effective from inception (the earliest origination, or 1900-01-01) with the same label; a test that every contract date resolves every setting
H11	Staging thresholds are resolved at today's date, not the period end, and sit outside maker-checker	StagingThreshold.php:36-46 uses now(); StagingService.php:113 silently defaults to [31,181]	A re-run of a past period uses today's thresholds, against D21; a missing row is not an error	Resolve at the period end; thresholds proposed and approved like every other setting; no silent default
H12	Mega Farm ECL misreads decision D30	MegaFarmEclService.php:58,117-125: ecl_value = EAD × PD × LGD × 5%	The loan allowance is booked at 5 percent of the programme loss. D30: the provision is the fund's; MAIIC's 5 percent is a share of interest, written down separately. Neither the fund-side provision nor the receivable write-down is produced; a later portfolio ECL run restores the loan to 100 percent (ExpectedCreditLossController.php:375 has no GL exclusion)	Awaits Dr Thom's confirmation of D30 as stated in the After Build Report; until then the Mega Farm figure must not be quoted
H13	The audit's own --verify covers 9 checks against 13 section-9 ties and 6 golden numbers	BaselineService.php	No 2024 year-end GL tie, no monthly GL tie, no interest-income YTD tie, no 28-posting count, no feed re-pull test, no collateral or take-on counts; the ECL golden numbers await the first approved run	Add the missing ties as the data for each is confirmed; the After Build Report's "9 checks, 0 FAIL" is accurate about what runs and should say so

#	Finding	Where	Effect	Fix
H1 4	The full test suite could not run in one process	Eleven legacy controllers call <code>ini_set('max_execution_time', 300)</code> for their own long requests (ExpectedCreditLossController.php:287, LoanBookController.php:562, CollateralController.php:324, the LGD controllers); once a test touches one, every test that follows shares a five-minute clock and the suite dies with "Maximum execution time exceeded" (three attempts, each from a different test)	A CI or release check that runs "the tests" silently runs part of them; the After Build Report's "308 passed" was a directory run, not the suite	Fixed: <code>tests/TestCase::setUp</code> resets the limit to unbounded before every test; the controllers should raise the limit only when it is lower (not lower it from unlimited)

4.3 Medium

#	Finding	Where
M1	Trial balances bypass the landing zone (imported directly, no hash, no load row); GL_02 is landed but not read by the reconciliation	Bootstrap.php:142-147; TrialBalanceImportService.php; EirGLReconciliationService.php
M2	Gates missing at the door: no account-in-master check, no balance-history-to-ledger tie, no month-end-row-per-account check, no numeric parse, query version recorded but not validated; quarantine keeps no rows; dates accepted in whatever format the manifest declares (the committed pack is m/d/yyyy, against D23)	PackLandingService.php:65-129,145-148,288-299
M3	The manual-overlay register of spec 14.6 does not exist; the "Manual overlay" route reads the latest legacy fli_adj row	FliRouteService.php:106-111
M4	The overlay-lock rule of 15.7 is an empty if block; lock() checks nothing	ScenarioSetService.php:91-93,154-162
M5	The Scenario Sets screen cannot set weights, add scenarios or edit shocks; only the fixed first set can be seeded. The decision the spec reserves for the CFO has no screen	ScenarioSetController.php; Pages/Governance/ScenarioSets.vue
M6	Two regression chains run in parallel: the new univariate path under maker-checker, and the legacy screen (RegressionController.php:227-233) with one-click approval, no test and no second person, still in the menu as "Regression Analysis"	
M7	Take-on "recompute from origination" writes a label and nothing else: no EIR solve from the workbook schedule, no TAKEON_WORKBOOK schedule	TakeonLandingService.php:179-228
M8	The bootstrap gives Governance-Centre-type approvals it is told never to give: fee classifications reviewed under the label, EIR locks by administrator override; and, inconsistently, leaves the 30 seeded fee rules unapproved, so 19 contracts stall at the fee gate and EIR coverage on the clean install is 12 of 135 loans	Bootstrap.php:182-189,242; FeeRuleMatcher.php:72-77
M9	Sensitivity reads coalesce(pd_prefli, pd_value, 12m_pd) where 12m_pd is percent from one writer and a fraction from another; and uses the stored ead column, which the build fills at 100 percent utilisation unlike every other EAD reader	ScenarioSetService.php:252-257; LoanBookBuildService.php:555; SeedReportFields.php:81
M10	PD is assigned by the pre-qualitative stage and the matrix is pre-to-pre; the instalment trigger is outside the PD's default definition; loans that leave the book count as "Paid" in the denominator	PdEngineService.php:67,69; MaiicTransitionProfileSeeder.php:22
M11	No cure or probation period: stage is recomputed from current DPD only; the prior stage is selected and never read	StagingService.php:53-71
M12	Fourteen catalogue settings are read by nothing but the Governance Centre and the trace (fli_expected_sign_test, macro_source_precedence, ebanker_feed_route and others); the FLI thresholds live in a second store (governed_parameters) with different seed defaults that a direct write would bypass	GovernanceService.php; GovernedValues.php:38-64; PollFeedFolder.php:37-41
M13	As-at date: CSV only (spec: Excel and PDF); part-month accrues on the EIR compounded rather than the section-3 convention; take-on basis not read	EirAsAtService.php:71-75; EirAsAtController.php:76

#	Finding	Where
M14	Zip extraction from the feed API and the poller relies on PHP's internal path sanitisation with no explicit check against <code>..</code> entries	<code>routes/web.php:1319</code> ; <code>PollFeedFolder.php</code>
M15	<code>COLLECTION_TYPES</code> still contains 'Interest' (spec 7.4 says corrected); reversal pairing (7.1) not found; over-sanction flag (7.3) not found; <code>RATE_BASIS</code> mapping still in the legacy Excel importer	<code>EirRevenueService.php:29</code> ; <code>ContractMasterImport.php:71-75</code>
M16	Help chapters still carry the old group names; Correlation Finder and Auditor Pack are CLI-only with no screen; <code>macro.view</code> / <code>macro.manage</code> permissions do not exist (macro commit uses <code>eir.govern</code>)	<code>help seeders 07a, 08</code> ; <code>config/menu.php</code> ; <code>MacroStatisticsController.php:26</code>
M17	Engine chain in the bootstrap runs PD, LGD, FLI, ECL, stress and reports for the last period only, against "every period from the first full month"	<code>Bootstrap.php:263-367</code>
M18	Section 16 beyond staging and PD is not built: the 15 percent interest split, the maize repayment and buyer receivable, the fund liability roll-forward, the Mega Farms report, seasonal segmentation; LGD borrowed from MAIIC's own book; scalar fallback hard-coded from the statements	<code>MegaFarmEclService.php:150-151,168-179</code>

4.4 Low and hygiene

#	Finding	Where
L1	The repository carries a 715-file corporate/ folder (a credit-scoring model with .exe, .rar, raw InnoDB .ibd and SQL dumps) and a medical ICD-10 XML under storage/app, both from the product's initial commit, both deployed to the client's server	corporate/; storage/app/icd10cm_tabular_2021.xml
L2	Other-client strings outside docs/reference: FDH in method keys and worked examples the user sees (AdjustmentMethods.php, TransmissionMethodCatalogue.php:181-183, GovernedValues.php:47, 54, Guardrail.php:18, StressTestingController.php:14-15); ZNBS in seeded support-ticket text (TicketSeeder.php:234, 253); three committed session transcripts under docs/sessions naming other clients throughout; build scripts under docs/build-files with other-client paths. The compliance PDF footer was the worst instance and is fixed at f5b08fc	
L3	Composer advisories (symfony/routing ×2 medium, polyfill-intl-idn low), two abandoned packages; npm 33 advisories (TinyMCE XSS in the help editor, yaml)	composer.lock, package-lock.json
L4	.env.example and config: cache prefix was shared across the production, demo and bootstrap env files (fixed 8 October: keyed by APP_ENV); SESSION_SECURE_COOKIE unset by default; dead CSRF exclusion file/upload	config/cache.php; config/session.php:171; VerifyCsrfToken.php:14
L5	ScheduleWorkflowService.php:76 treats a stated rate above 1 as a percent, so a stated "1" (1 percent) becomes 100 percent; CalculateEirService.php:88 30/360 is 30E/360; lifetime_pd and 12m_pd are decimal(8,2) so an 8-dp write is truncated; StagingService.php:156 casts TRANSAMT without stripping commas	
L6	Legacy test suites (92 "medical" tests and five FLI page tests) fail on master before this engagement and remain in the tree; test classes build their own sqlite schemas by hand, so the real migrations are exercised only by the bootstrap	tests/

5. Coverage of the EIR chain on a clean install

The After Build Report quotes "31 schedules, 31 EIRs solved, 12 locked". The audit traced the other 113 contracts and the 19 unlocked EIRs on the freshly bootstrapped database:

Reason the schedule was not generated	Contract s	Whose it is
A moratorium is stated but its type is not, on the contract or its scheme	51	MAIIC (a decision on the default moratorium type, or the scheme data)
Drawn amount not positive (with other reasons)	39	Data: undisbursed or held contracts; the readiness rule is right to refuse them
No approved value for day_count at the origination date	16	Dupleix (finding H10)
Part-drawn and neither contract nor scheme says whether interest is on the drawn amount	5	MAIIC (partly_drawn_interest_basis is a catalogue setting nothing reads yet; M12)

Reason the schedule was not generated	Contract s	Whose it is
Contractual rate missing or invalid	2	Data
EIR solved but not locked: fee lines not independently reviewed	19	The 30 seeded fee rules are unapproved by design (Dr Thom's rulebook approval is owed); see M8 for the bootstrap's inconsistency

So on today's clean install the revenue roll-forward, the as-at service and the GL reconciliation cover 12 of 135 loans; the ECL covers all 135. Both facts belong in the After Build Report's section 3 in those words.

6. Corrected during the audit

Each is committed with its evidence.

Commit	What
f5b08fc	RBM bands to the Gazette (C4) in the return, the classification report, the provision comparison, the arrears ageing and the test; compliance PDF footer names MAIIC
f324814 (8 October, found by the dark-mode sweep's capture of every screen)	Six ECL-over-EAD ratios divided by the string "0.00" (the sector and product-group ECL reports threw on a sector with no exposure); the report smoke now covers all 30 report methods; the licence page on a fresh database; the cache prefix keyed by environment (a permission cache filled by one database was served to another)
742e7a7 (8 October)	The RBM Classification report put on one rule with the return (the rule itself then corrected at f5b08fc)

7. Test suite

The whole suite was run in one process for the first time (it had died at five minutes on every earlier attempt; see H14, fixed). Result: **577 tests, 2,779 assertions, 6 minutes 33 seconds; 455 passed, 30 skipped, 71 errors, 21 failures.**

Failing group	Count	Cause	Belongs to
Medical and inventory modules (patients, vitals, invoices, inventory purchases and sales, tariffs, SMS gateways, expense types, claim batches, appointments)	79	Models, factories and the receptionist role no longer exist; the tests are left over from the product the platform was forked from	The fork; retire the tests and the modules
Legacy FLI page tests (Tests\Feature\FLI\ExternalCalculationsTest)	5	403 on the legacy external-calculations screen and a NOT NULL constraint in its parameters table; failing on master before this engagement	The legacy FLI chain (M6)
Jetstream scaffold (RegistrationTest, ProfileInformationTest, DeleteAccountTest), DatabaseTest, LoanApplicationApprovalStageAssignedMailTest	8	Features disabled or models removed (App\Models\Role not found; users table absent from the hand-built sqlite schema)	The fork
Tests written in this engagement (EIR, E-Banker, staging, PD/LGD, FLI chain and route, scenario sets, compliance, Mega Farm, RBM, navigation, help)	0 failing		

Two observations on the suite itself. First, the engagement's tests build their own sqlite schemas by hand, so the real MySQL migrations are exercised only by the bootstrap (which does run them from nothing, so they are proven, but a column drift between a test schema and a migration would not be caught by the tests). Second, several engine tests encode the behaviour the engine has rather than the behaviour IFRS 9 requires

(C3's covering test asserts the understated range; C4's test asserted the wrong bands until today): a passing test is evidence of reproduction, not of correctness, and the fixes in section 9 must change the expectations, not just the code.

8. What this means for the After Build Report's figures

The After Build Report of 8 October stands as a record of what runs and what ties. The following figures in it should not be quoted to the board, the auditor or the Reserve Bank until the findings against them are closed:

Figure	Rests on
ECL 8.94bn post-FLI (8.33bn pre-FLI) at August 2026	C1 (12-month PD for Stage 2, undiscounted), C9 (a fit the bootstrap approved), H1 (300/401 as cash in the carrying amounts), H5 (stage split in the reports)
Weighted sensitivity 10.24bn; stress severe 11.86bn	H3 (two weightings, two scenario sets) on top of the above
Revenue roll-forward and Stage 3 net interest	C5, C6; and coverage of 12 of 135 loans (section 5)
Mega Farm 23.5bn programme ECL (demo)	H12 (D30 misread); already marked as awaiting confirmation
RBM return and classification	Now on the Gazette's bands (C4 fixed); the rates and sections were right; the prescribed form is still awaited
The nine golden ties	Sound; they prove the ledger reproduction and the GL ties, which H1 does not disturb because the stored report makes the same classification

9. Order of work

What Duplex should do, in this order, before the next figure goes to MAIIC. Effort is an estimate for one engineer.

- Authenticate the legacy modules and bind the SQL** (C7, C8): a day. Nothing else matters if the server is reachable first.
- Stage 2 lifetime PD, one tenor unit, Stage 3 from the reporting date, discounted LGD** (C1, C2, C3, H7): three to four days, with a worked example per stage in the tests and the FS note re-run.
- Stage 3 allowance into the revenue engine; revenue after ECL; schedule fallback bounded by the feed** (C5, C6): two days; the golden revenue-shift number is born here, so Dr Thom's first approved run follows this.
- 300 and 401 out of cash** (H1): a day plus the re-proof of the ties with the exceptions stated.
- Bootstrap to the spec's rule on fits and approvals; one scenario weighting; the sensitivity from the per-scenario ECL** (C9, H3, M8): two days.
- Governance defaults from inception; thresholds at the period end under maker-checker; the lock honoured by every writer; admin override governed** (H10, H11, H4, H9): two days.
- Tranche drawdowns in the EIR vector; PD annualisation; post-stage splits; gross carrying in the note** (H2, H8, H5, H6): three days.
- The remaining spec gaps** (M1 to M18): the overlay register, the set editor, the gates, the trial-balance landing, the legacy regression chain, section 16 once D30 is confirmed: two to three weeks, to be sequenced with MAIIC's decisions.
- Hygiene** (L1 to L6): remove corporate/ and the ICD file, strip other-client strings, update dependencies, retire the legacy suites, lift the execution limit: a day.

Decisions MAIIC owes that the audit makes more urgent: D30 (Mega Farm basis), the fee rulebook approval (19 EIRs wait on it), the default moratorium type (51 schedules wait on it), the partly-drawn interest basis (5), the prescribed RBM return form, and the first approved run from which the ECL and revenue-shift

golden numbers are born.

10. Files

Audit evidence: this report; the bootstrap log and the PHPUnit log in the engagement file; the four review reports (conformance, engine, security, operations) in the engagement file; the Gazette at docs/regulatory. Corrections: commits 742e7a7, f324814, f5b08fc on eir_revenue_recognition (master fast-forwarded).